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Data Science Intern

Week 2

Task: Sales Data Analysis Dashboard

Description: Analyze a sales dataset and create visual reports showing monthly sales trends, best-selling products, and customer insights.

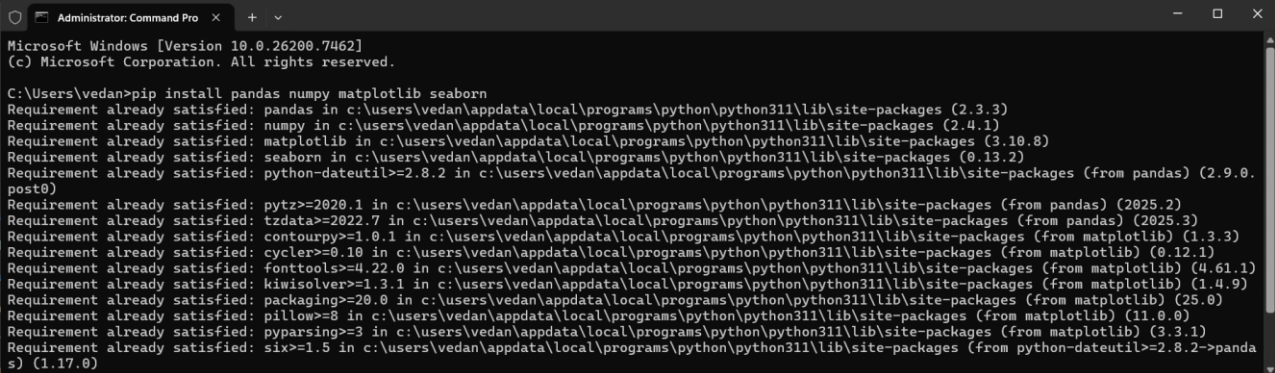
Challenges:

- Handle missing & inconsistent data
- Group by region/category for insights
- Create 3–5 meaningful visualizations

Tech Stack: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook.

Attached are the sequential screenshots and execution

- Installing the required libraries



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vedan>pip install pandas numpy matplotlib seaborn
Requirement already satisfied: pandas in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (2.3.3)
Requirement already satisfied: numpy in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (2.4.1)
Requirement already satisfied: matplotlib in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (3.10.8)
Requirement already satisfied: seaborn in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (11.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (3.3.1)
Requirement already satisfied: six>=1.5 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

(All the required libraries were already installed)

- Creating a new Jupyter notebook

```

C:\Users\vedan>npm prefix
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vedan>jupyter notebook

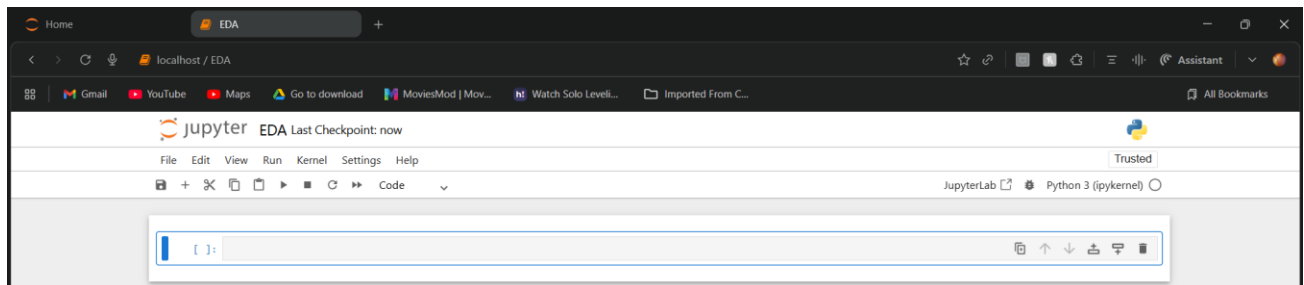
[1 2026-01-11 14:11:31.888 ServerApp] jupyter_lsp | extension was successfully linked.
[1 2026-01-11 14:11:31.894 ServerApp] jupyter_server_terminals | extension was successfully linked.
[1 2026-01-11 14:11:31.198 ServerApp] jupyterlab | extension was successfully linked.
[1 2026-01-11 14:11:31.185 ServerApp] notebook | extension was successfully linked.
[1 2026-01-11 14:11:31.386 ServerApp] notebook_shim | extension was successfully linked.
[1 2026-01-11 14:11:31.468 ServerApp] notebook_shim | extension was successfully loaded.
[1 2026-01-11 14:11:31.478 ServerApp] jupyter_lsp | extension was successfully loaded.
[1 2026-01-11 14:11:31.471 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[1 2026-01-11 14:11:31.481 LabApp] JupyterLab extension loaded from C:\Users\vedan\AppData\Local\Programs\Python\Python311\Lib\site-packages\jupyterlab
[1 2026-01-11 14:11:31.481 LabApp] JupyterLab application directory is C:\Users\vedan\AppData\Local\Programs\Python\Python311\share\jupyter\lab
[1 2026-01-11 14:11:31.482 LabApp] Extension Manager is 'pypi'.
[1 2026-01-11 14:11:31.749 ServerApp] jupyterlab | extension was successfully loaded.
[1 2026-01-11 14:11:31.753 ServerApp] notebook | extension was successfully loaded.
[1 2026-01-11 14:11:31.767 ServerApp] Serving notebooks from local directory: C:\Users\vedan
[1 2026-01-11 14:11:31.767 ServerApp] Jupyter Server 2.17.0 is running at:
[1 2026-01-11 14:11:31.768 ServerApp] http://localhost:8888/tree?token=5090540900f4243296f030e0fbf21b3aeb7010de27a383d
[1 2026-01-11 14:11:31.768 ServerApp] http://127.0.0.1:8888/tree?token=5090540900f4243296f030e0fbf21b3aeb7010de27a383d
[1 2026-01-11 14:11:31.768 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).

[C 2026-01-11 14:11:31.809 ServerApp]

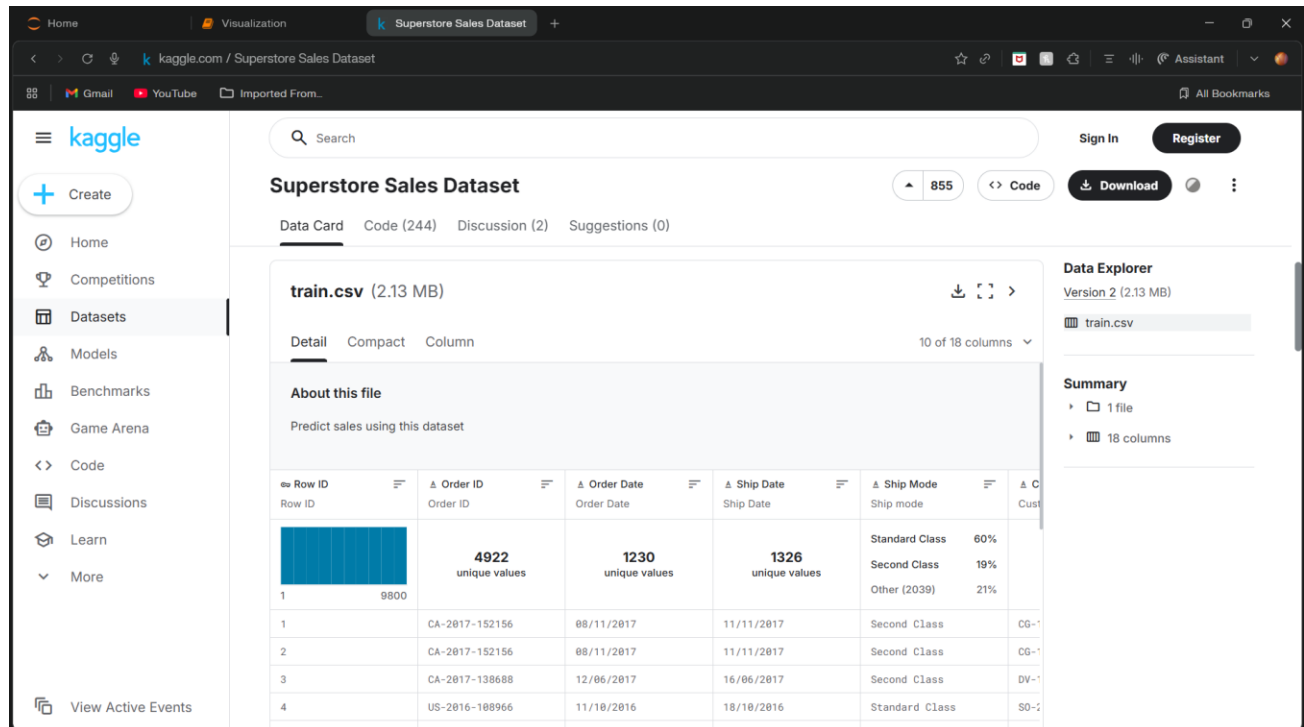
To access the server, open this file in a browser:
file:///C:/Users/vedan/AppData/Roaming/jupyter/runtime/jpserver-18316-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=5090540900f4243296f030e0fbf21b3aeb7010de27a383d
http://127.0.0.1:8888/tree?token=5090540900f4243296f030e0fbf21b3aeb7010de27a383d

[1 2026-01-11 14:11:32.377 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-type
script-langsver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, r-language-server, sql-language
-server, texlab, typescript-language-server, unified-language-server, vscode-css-language-server-bin, vscode-html-language-server-bin, vscode-json-language-server
ver-bin, yaml-language-server
[1 2026-01-11 14:11:48.625 ServerApp] Kernel started: 011a9c09-73e2-463c-8f24-4a9bed0afa6b
[1 2026-01-11 14:11:41.667 ServerApp] Adapting from protocol version 5.3 (kernel 011a9c09-73e2-463c-8f24-4a9bed0afa6b) to 5.4 (client).
[1 2026-01-11 14:11:41.667 ServerApp] Adapting from protocol version 5.3 (kernel 011a9c09-73e2-463c-8f24-4a9bed0afa6b) to 5.4 (client).
[E 2026-01-11 14:11:41.668 ServerApp] Uncaught exception GET /api/kernels/011a9c09-73e2-463c-8f24-4a9bed0afa6b/channels?session_id=69dfa832-fccb-443d-8c1c-d
ba3673948a5f (:::1)
HTTPServerRequest(protocol='http', host='localhost:8888', method='GET', uri='/api/kernels/011a9c09-73e2-463c-8f24-4a9bed0afa6b/channels?session_id=69dfa
832-fccb-443d-8c1c-dba3673948a5f', version='HTTP/1.1', remote_ip '::1')
Traceback (most recent call last):
File "C:\Users\vedan\AppData\Local\Programs\Python\Python311\Lib\site-packages\tornado\web.py", line 1859, in _execute
result = await result
          ^^^^^^^^^^^

```



- Download the dataset from Kaggle (Superstore Sales Dataset) (.csv files)



Superstore Sales Dataset (2.13 MB)

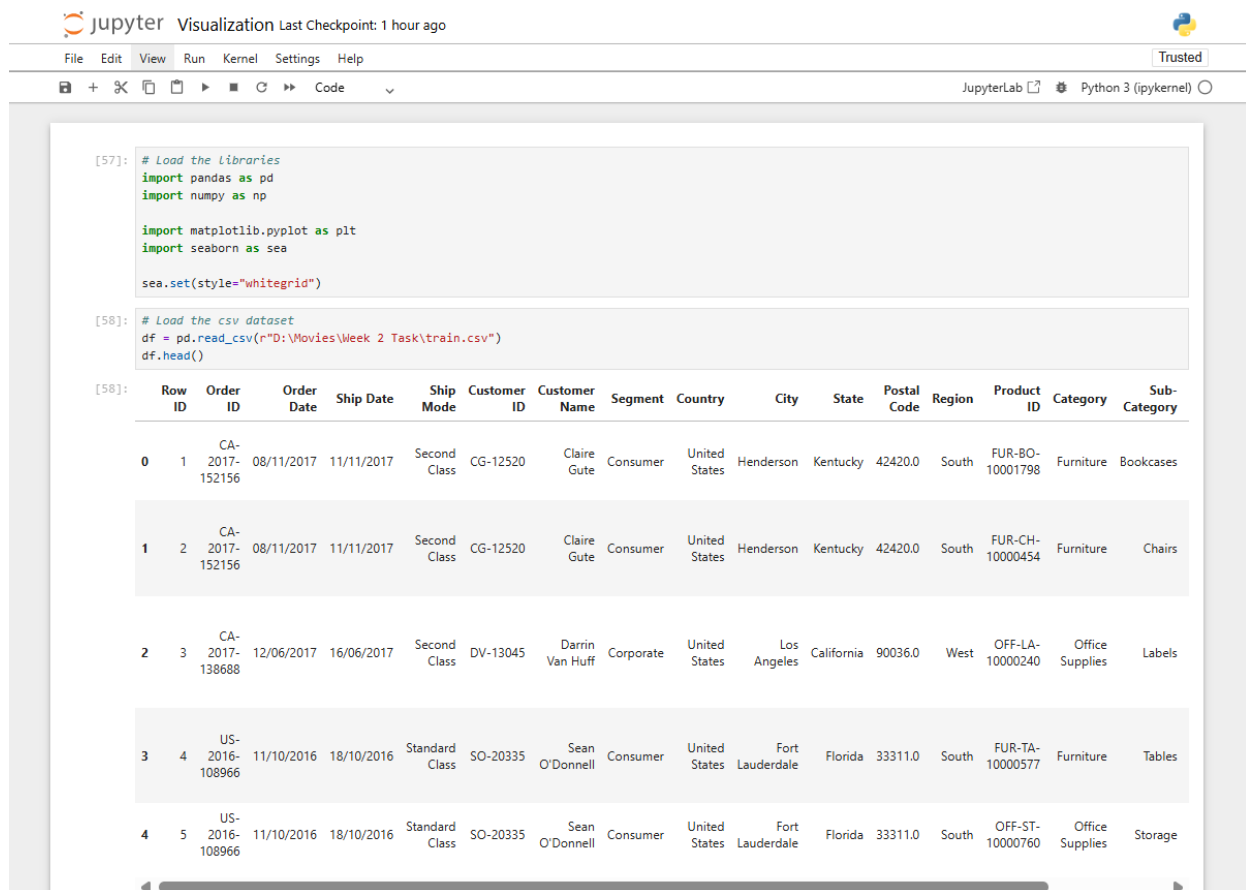
855 rows, 18 columns

Summary

- 1 file
- 18 columns

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gulte	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	Bookcases
2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gulte	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	Chairs
3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	Labels
4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577	Furniture	Tables
5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760	Office Supplies	Storage

- Importing libraries into the Notebook and viewing the Titanic Dataset



Jupyter Visualization Last Checkpoint: 1 hour ago

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JupyterLab Python 3 (ipykernel)

```
[57]: # Load the libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sea

sea.set(style="whitegrid")

[58]: # Load the csv dataset
df = pd.read_csv(r"D:\Movies\Week 2 Task\train.csv")
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gulte	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	Bookcases
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gulte	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	Chairs
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	Labels
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577	Furniture	Tables
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760	Office Supplies	Storage

Cell wise Jupyter Notebook Code : (Visualization.ipynb)

(The code snippets include a comment at the beginning showing the purpose)

```
# Load the Libraries
```

```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sea
```

```
sea.set(style="whitegrid")
```

```
# Load the csv dataset
```

```
df = pd.read_csv(r"D:\Movies\Week 2 Task\train.csv")
df.head()
```

```
# Label of the columns
```

```
print("Columns in dataset:")
print(df.columns)
```

```
# Physical View
```

```
df.info()
```

```
# Missing values and Duplicates
```

```
print("Missing values per column:\n", df.isnull().sum())
print("\nDuplicate rows: ", df.duplicated().sum())
```

```
# Convert Date columns
```

```
df_clean["Order Date"] = pd.to_datetime(df_clean["Order Date"], dayfirst=True, errors="coerce")
df_clean["Ship Date"] = pd.to_datetime(df_clean["Ship Date"], dayfirst=True, errors="coerce")
```

```
df_clean[["Order Date", "Ship Date"]].head()
```

```
# Separating Month and Year for Visualizations wrt Months
```

```
df_clean["Month"] = df_clean["Order Date"].dt.to_period("M")
df_clean[["Order Date", "Month"]].head()
```

```
# Monthly Sales
```

```
monthly_sales = df_clean.groupby("Month")["Sales"].sum()
monthly_sales.plot(kind="line", marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=90)
plt.grid(True)
plt.show()
```

```
# Top 10 Products by Sales
```

```
top_products = df_clean.groupby("Product Name")["Sales"].sum().sort_values(ascending=False).head(10)
top_products.plot(kind="bar")
plt.title("Top 10 Best Selling Products")
plt.xlabel("Product Name")
plt.ylabel("Total Sales")
plt.xticks(rotation=90)
plt.show()
```

```
# Region Wise Sales
```

```
region_sales = df_clean.groupby("Region")["Sales"].sum().sort_values(ascending=False)
region_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=90)
plt.show()
```

```
# Top 10 Customers
```

```
top_customers = df_clean.groupby("Customer Name")["Sales"].sum().sort_values(ascending=False).head(10)
top_customers.plot(kind="bar")
plt.title("Top 10 Customers by Sales")
plt.xlabel("Customer Name")
plt.ylabel("Total Sales")
plt.xticks(rotation=90)
plt.show()
```

```
# Final Summary
print("Sales Summary")
print("\n")
print("Dataset Shape:", df_clean.shape)
print("Total Sales:", df_clean["Sales"].sum())
print("Average Sales:", df_clean["Sales"].mean())

print("\nTop 5 Products:")
print(df_clean.groupby("Product Name")["Sales"].sum().sort_values(ascending=False).head(5))

print("\nTop Region:")
print(df_clean.groupby("Region")["Sales"].sum().sort_values(ascending=False).head(1))

print("\nTop Category:")
print(df_clean.groupby("Category")["Sales"].sum().sort_values(ascending=False).head(1))
```

Output Snippets :

The JupyterLab interface displays the following code and output:

```
[57]: # Load the libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sea
sea.set(style="whitegrid")

[58]: # Load the csv dataset
df = pd.read_csv("D:\Movies\Week 2 Task\train.csv")
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	Bookcases
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	Chairs
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	Labels

The JupyterLab interface displays the following code and output:

```
[59]: # Label of the columns
print("Columns in dataset:")
print(df.columns)

[60]: # Physical View
df.info()

[61]: # Missing values and Duplicates
print("Missing values per column:\n", df.isnull().sum())
print("\nDuplicate rows: ", df.duplicated().sum())
```

Columns in dataset:
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales'], dtype='object')

```
Missing values per column:
Row ID      0
Order ID    0
```

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JupyterLab Python 3 (ipykernel)

```
[61]: # Missing values and Duplicates
print("Missing values per column:\n", df.isnull().sum())
print("\nDuplicate rows: ", df.duplicated().sum())

Missing values per column:
Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID  0
Customer Name 0
Segment     0
Country     0
City        0
State       0
Postal Code 11
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
dtype: int64

Duplicate rows: 0

[62]: # Convert Date columns
df_clean["Order Date"] = pd.to_datetime(df_clean["Order Date"], dayfirst=True, errors="coerce")
df_clean["Ship Date"] = pd.to_datetime(df_clean["Ship Date"], dayfirst=True, errors="coerce")

df_clean[["Order Date", "Ship Date"]].head()
```

	Order Date	Ship Date
0	2017-11-08	2017-11-11

Jupyter Visualization Last Checkpoint: 1 hour ago

File Edit View Run Kernel Settings Help

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JupyterLab Python 3 (ipykernel)

```
df_clean[["Order Date", "Ship Date"]].head()

[62]:
```

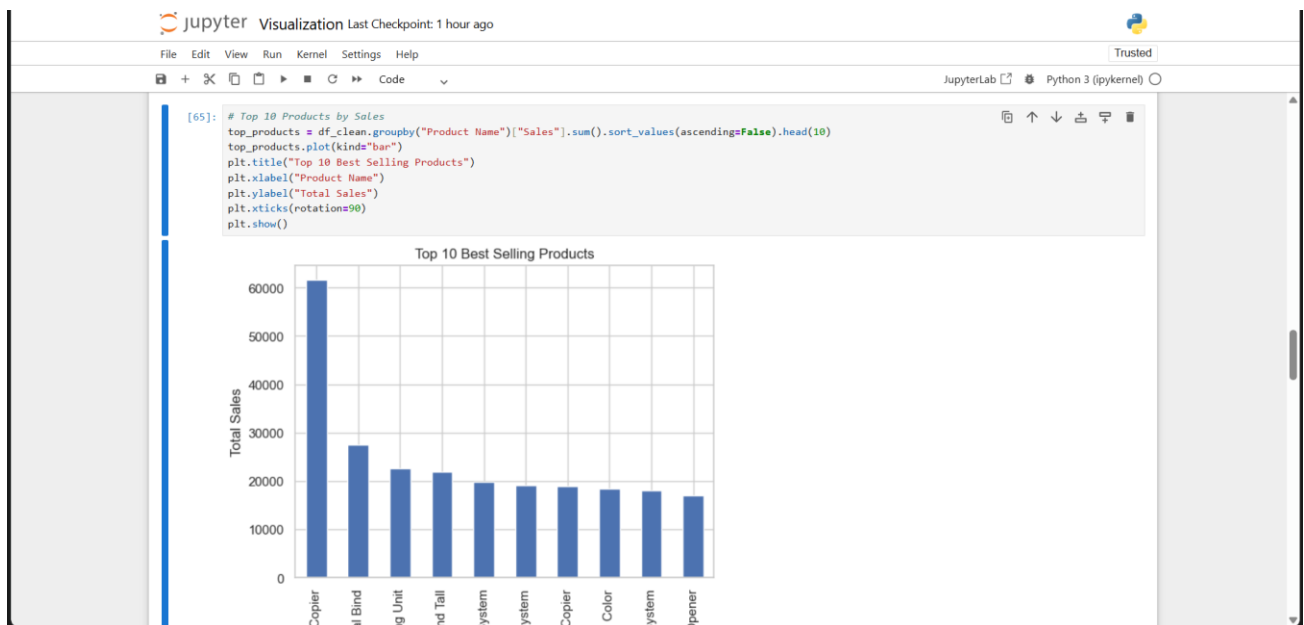
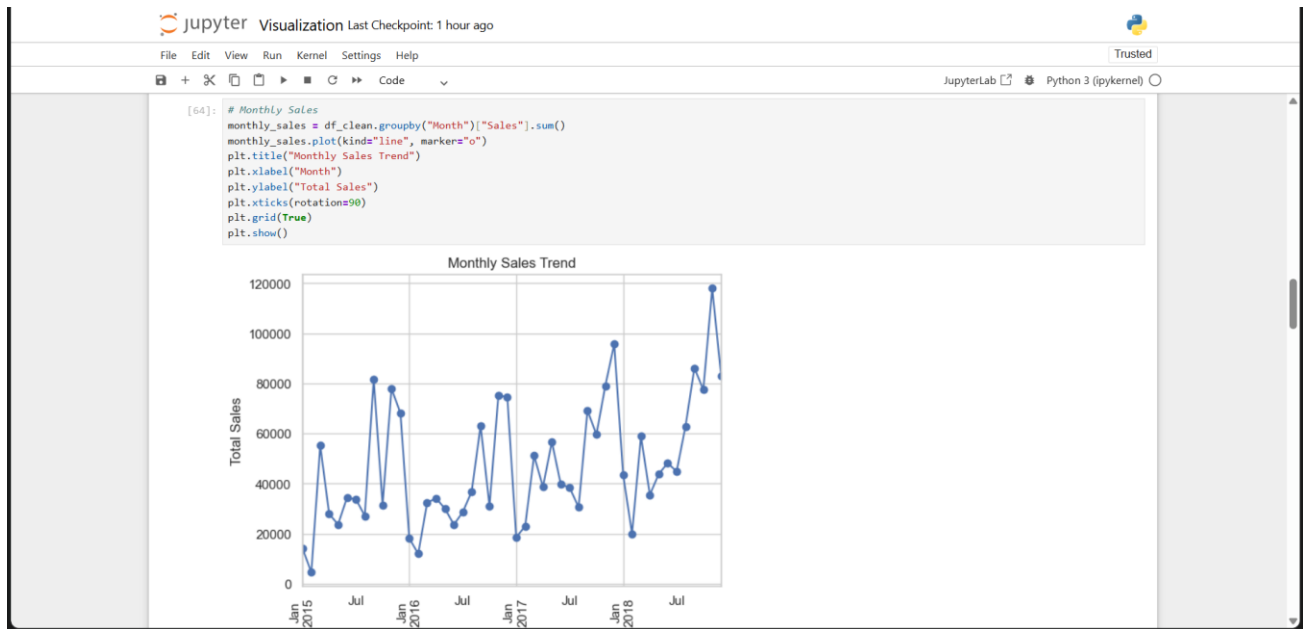
	Order Date	Ship Date
0	2017-11-08	2017-11-11
1	2017-11-08	2017-11-11
2	2017-06-12	2017-06-16
3	2016-10-11	2016-10-18
4	2016-10-11	2016-10-18

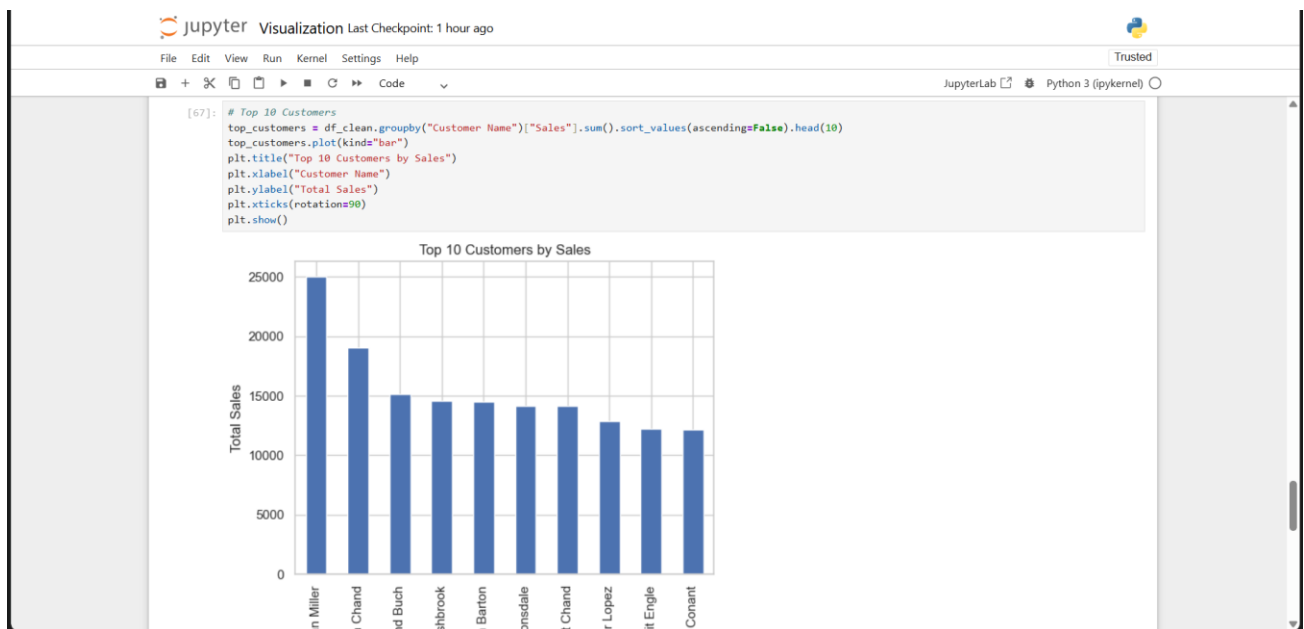
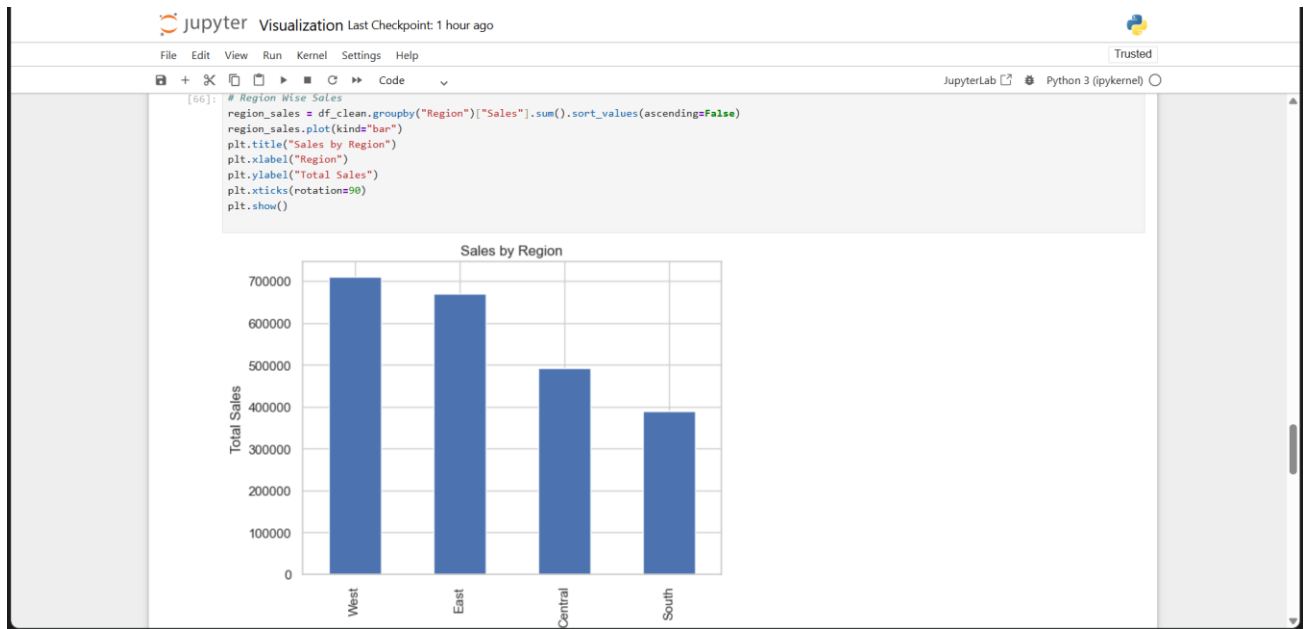
```
[63]: # Separating Month and Year for Visualizations wrt Months
df_clean["Month"] = df_clean["Order Date"].dt.to_period("M")
df_clean[["Order Date", "Month"]].head()

[63]:
```

	Order Date	Month
0	2017-11-08	2017-11
1	2017-11-08	2017-11
2	2017-06-12	2017-06
3	2016-10-11	2016-10
4	2016-10-11	2016-10

```
[64]: # Monthly Sales
monthly_sales = df_clean.groupby("Month")["Sales"].sum()
monthly_sales.plot(kind="line", marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=90)
```






```
[68]: # Final Summary
print("Sales Summary")
print("\n")
print("Dataset Shape:", df_clean.shape)
print("Total Sales:", df_clean["Sales"].sum())
print("Average Sales:", df_clean["Sales"].mean())

print("\nTop 5 Products:")
print(df_clean.groupby("Product Name")["Sales"].sum().sort_values(ascending=False).head(5))

print("\nTop Region:")
print(df_clean.groupby("Region")["Sales"].sum().sort_values(ascending=False).head(1))

print("\nTop Category:")
print(df_clean.groupby("Category")["Sales"].sum().sort_values(ascending=False).head(1))
```

Sales Summary

Dataset Shape: (9800, 19)
Total Sales: 2261536.7827
Average Sales: 230.76905945918367

Top 5 Products:

Product Name	Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479

Name: Sales, dtype: float64

Top Region:

Region	Sales
West	718219.6845

Name: Sales, dtype: float64

Top Category:

Category	Sales
Technology	827455.873

Name: Sales, dtype: float64